# TRIANGLE FRAMES AND ROOT-SPARSE LATTICES FOR CONWAY'S 99-GRAPH

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**ABSTRACT**

We obtain necessary conditions for a strongly regular graph with parameters $(99, 14, 1, 2)$, without assuming any automorphisms. Let $N_m$ count unordered pairs of disjoint triangles joined by exactly $m$ edges, and put $\delta = N_2 - 693$. A cubic tensor associated with an integral triangle frame, together with a local lattice norm gap, gives $\delta = 3465 - 3N_3 \in 3\mathbb{Z}$ and $\delta \geq 165$. Thus $N_2 \geq 858$ and $N_3 \leq 1100$. We also construct an even positive-definite lattice of rank $44$ and determinant $49$ containing the triangle lattice. Its dual has minimum squared norm at least $2$, and its root system is $A_1^\rho$ with $0 \leq \rho \leq 3$. The root bound comes from an interpretation of roots by Hoffman equality cocliques and a combinatorial obstruction to four directions. We give proofs not requiring exhaustive enumeration and provide two complete Lean formalizations of the main theorem. These results do not decide the existence of Conway's 99-graph. The mathematical research was conducted largely autonomously by AI agents under human direction; their roles and the validation protocol are disclosed in Appendix C.

***Keywords*** strongly regular graphs · Conway's 99-graph · integral lattices · tight frames · Hoffman bound · formal verification

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## 1 Introduction and main result

The existence question goes back to Norman Biggs in 1969, as recalled in the historical account of Cesarz and Woldar [8], and was later popularized by Conway's prize problems [1]. It provides a longstanding, concrete test case for the gap between feasible strongly regular graph parameters and actual graph realizability.

We consider a simple undirected strongly regular graph $G$ with parameters $(99, 14, 1, 2)$ and adjacency matrix $A$. Thus every vertex has $14$ neighbours, adjacent vertices have one common neighbour, and distinct non-adjacent vertices have two common neighbours. The purpose of this paper is to derive necessary conditions by combining triangle counts, integral tight frames, and lattice geometry. No construction or proof of non-existence is obtained. Conway's formulation is recorded in [1]; general background on strongly regular graphs can be found in [2].

Write $\mathcal{T}$ for the set of triangles of $G$. For $m \in \{0, 1, 2, 3\}$, let $N_m$ denote the number of *unordered* pairs of disjoint triangles joined by exactly $m$ edges. Set

$$\delta = N_2 - 693.$$

**Theorem 1.1.** *Every strongly regular graph with parameters* $(99, 14, 1, 2)$ *satisfies*

$$N_2 + 3N_3 = 4158, \qquad \delta = 3465 - 3N_3 \in 3\mathbb{Z}, \qquad \delta \geq 165.$$

*In particular,* $N_2 \geq 858$ *and* $N_3 \leq 1100$*.*

*Moreover, the graph determines a triangle lattice* $\Lambda$ *contained in an even positive-definite integral lattice* $\widehat{N}$ *of rank* $44$ *and determinant* $49$*, such that*

$$\min(\widehat{N}^*) \geq 2, \qquad \Phi(\widehat{N}) = A_1^\rho, \qquad 0 \leq \rho \leq 3.$$

*Here the minimum is taken over* squared norms *of nonzero vectors. For an integral lattice* $M$*,* $\Phi(M)$ *denotes its vectors of squared norm* $2$*. The notation* $A_1^\rho$ *means that its* $\rho$ *root directions are pairwise orthogonal.*

In fact the root bound is uniform: every even integral overlattice of the canonical lattice $N$ defined in (14) has root system $A_1^\rho$ with $\rho \le 3$. The determinant-49 construction is not needed for that assertion. The choice of $\widehat{N}$ need not be unique.

### 1.1 Relation to earlier work

The problem has an extensive literature, and its standard spectral and counting preliminaries should not be confused with the contributions here. Wilbrink [3] excluded automorphisms of order 11. Automorphism restrictions were subsequently developed by Makhnev and Minakova [5], Behbahani and Lam [6], and Crnković and Maksimović [7]. Cesarz and Woldar [8] give recent computer-free restrictions on the automorphism group. These results concern possible symmetry; all the conditions in Theorem 1.1 hold without a symmetry hypothesis. Makhnev's work on graphs with $\lambda = 1$ [4] is also part of the earlier structural literature.

Our triangle-pair counts belong to an existing framework. Lou and Murin [9, Section 5] study the triangle-intersection graph and obtain fixed-triangle identities whose sum is precisely $N_2 + 3N_3 = 4158$. Their Theorem 4.3 also gives the independence bound 22 and the equality condition used below. Reimbayev [10, 11] relates six-vertex configurations to induced hexagons. In his notation, our $(N_3, N_2, N_1, N_0)$ correspond to $(n_1, n_3, n_5, n_{14})$, respectively. Petro and Phillips [12] develop the clique-graph viewpoint, including its application to these parameters. The spectral representation and Hoffman inequality used here are classical [13].

Arithmetic restrictions are not in themselves new in this problem: Ducey et al. [14] determine the critical group forced by the parameters. That Laplacian invariant is different from the discriminant form of the rank-44 lattice studied here. Our passage from this discriminant form to an overlattice uses classical theory, in particular Milgram's formula and isotropic gluing [15, 17].

Recent computational approaches provide complementary context. Keramatipour [18] investigates SAT encodings of the existence problem and their computational limitations. Thakkar and Severini [19] combine a forced-structure reduction with CP-SAT formulations and partial-construction search, including a search heuristic produced by an evolutionary program-search system. Their computational bounds and partial scores have a different scope from our unconditional triangle-count and lattice constraints; we do not interpret partial scores as bounds on $\delta$.

The contributions established in this paper are the local integral-tensor obstruction giving $\delta \ge 165$, and the graph-specific combination of an elementary discriminant at 7, a determinant-49 extension, a dual norm bound, and a root-rank bound obtained through Hoffman cocliques. We claim neither that the preliminary identities are new nor that every earlier necessary condition is subsumed by these results. The comparison above identifies the sources used and the mathematical distinctions; it is not an exhaustive priority claim.

*Remark* 1.2 (An induced-hexagon consequence). If $p_6$ denotes the number of induced hexagons, Reimbayev's identity [10, Theorem 3.3], in the normalization just specified, gives

$$p_6 = 209286 + N_2 = 209979 + \delta.$$

Consequently Theorem 1.1 implies $p_6 \ge 210144$. This is a consequence of a previously established counting identity, not a separate new counting method. The identity for $p_6$ is not among the statements verified by our Lean developments.

The triangle frame is introduced in Section 3. The proof of $\delta \ge 165$ in Section 4 is independent of the auxiliary lattice and Milgram's formula. Sections 5–7 construct $\widehat{N}$, describe its roots, and establish $\rho \le 4$. Sections 8–10 exclude $\rho = 4$ by two combinatorial arguments. No exhaustive search is required in the written proof. Appendix A proves the form of Milgram's identity used here. Two complete Lean developments of Theorem 1.1 are described in Appendix B.

Throughout, $w_u = 3y_u$ and $x_L = \sum_{u \in L} y_u$. The symbol $\rho$ always refers to the root rank of $\widehat{N}$. The symbol $\mathbf{1}$ denotes the all-ones column vector of the appropriate dimension, and $\mathbf{1}_H$ denotes the indicator vector of a set $H$.

## 2 Overview of the methods and their mathematical significance

Recent discussions by Tao and Cohn emphasize that the value of AI-generated mathematics depends not only on establishing results, but also on communicating the ideas that make them understandable and reusable [22, 23]. With this distinction in mind, we explain the mechanisms behind our proofs and how they connect the combinatorics, spectral geometry, and lattice structure of a hypothetical Conway graph.

There are two complementary arguments. The first measures a triangle-count defect by a cubic tensor and forces it to be positive by integrality. The second converts lattice roots into extremal independent sets and excludes four root directions by a small combinatorial budget. Figure 1 separates their dependencies.

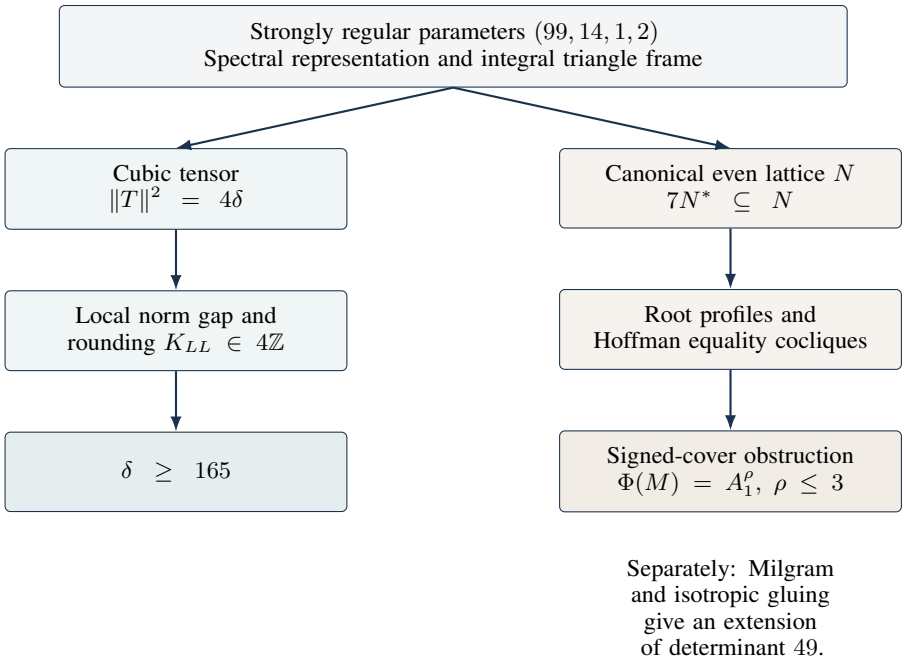



Figure 1: Two branches of the proof. The right-hand root bound holds for every even integral overlattice $M$ of $N$; it does not require the determinant-49 extension.

### 2.1 A count defect as the energy of a cubic tensor

Projecting vertices onto the $-4$ eigenspace is standard. The useful normalization here makes the sums over triangles integral: the 231 triangle vectors $x_L$ have squared norm 4, pairings in $\{0, 1, -1, -2\}$ off the diagonal, and satisfy $\sum_L x_L x_L^T = 21I$. Their cubic moment $T = \sum_L x_L^{\otimes 3}$ satisfies

$$\|T\|^2 = 4\delta.$$

Thus a combinatorial defect is exactly a nonnegative geometric energy, not merely bounded by one. This alone gives only $\delta \geq 0$.

The extra information is local and arithmetic. Contracting $T$ with a vertex vector produces a self-adjoint operator $M_u$ preserving the triangle lattice and fixing that vertex vector. Every image $M_u x_L$ has a specified nonzero residue against $w_u$. The same residue is impossible for a root. Evenness therefore raises its squared norm from the possible value 2 to at least 4. For each fixed triangle the tight frame converts these 99 norm gaps into $7K_{LL} \geq 396$. Crucially, $K_{LL}$ is a multiple of 4, so it must be at least 60. Summing only after this local rounding gives $84\delta \geq 231 \cdot 60$.

The conceptual point is the order of operations: preserve the integral congruence until after the local estimate, rather than average it away. The proof does not classify root systems, assume a group action, or enumerate graphs. The tensor is harmonic, but its harmonicity is not a substitute for the congruence and norm-gap argument.

### 2.2 A small discriminant, without a lattice classification

The triangle lattice sits inside a canonical even lattice $N$ generated by scaled vertices and their differences. A dual vector is described by an integral vertex profile whose entries share a residue modulo 3. Subtracting that common residue proves $7N^* \subseteq N$ directly. Hence the discriminant form is a quadratic space over $\mathbb{F}_7$. Milgram's classical signature identity forces its normalized Gauss sum to be $-1$. Splitting off hyperbolic planes leaves an anisotropic plane; gluing the isotropic part produces determinant $7^2$.

The input is graph-specific; the discriminant and gluing machinery is classical. In particular, we need no classification of positive-definite lattices of rank 44. The resulting extension need not be canonical.

### 2.3 Roots become extremal cocliques

Short dual vectors give integral $-4$ eigenvectors with entries $0, \pm 1$. Equality in the norm bound identifies their two seven-point supports as biplanes. The common-neighbour restriction then excludes nonorthogonal root directions.

Each root supplies a pair of 22-vertex independent sets attaining Hoffman's bound. Integrality severely restricts how such a set can interact with another root. For orthogonal roots, the signed supports meet in four singleton cells. A Gram matrix of their centres first gives at most four directions.

Four directions would leave 39 vertices outside four disjoint 15-point sets. Their signatures have fixed first and second weight moments. A zero signature has only two remaining neighbours and forces both to have weight four, leaving at most one zero signature. If there is none, a five-neighbour weight budget excludes every weight-three signature and contradicts the moments. If there is one, six integral profiles exhaust a twelve-edge budget. The surviving double signatures form a forced twelve-arc configuration, which supplies three common neighbours to a pair of vertices. This contradicts the parameters.

The result is a root-sparsity obstruction, not a nonexistence theorem: an even lattice may have no roots at all. Both this obstruction and the tensor bound are unconditional necessary conditions for the graph, but neither establishes a converse realization theorem.

## 3 Counts, Euclidean representation, and the cubic tensor

### 3.1 Triangles and disjoint pairs

We include these counts to fix normalizations. The fixed-triangle framework of Lou–Murin [9, Section 5] already yields the global identity (2).

Every edge lies in exactly one triangle, since $\lambda = 1$. The neighbourhood of a vertex is a disjoint union of seven edges: every vertex in that neighbourhood has exactly one neighbour there. There are $693 = 99 \cdot 14/2$ edges and $231 = 693/3$ triangles, and every vertex lies in seven triangles.

The edges between two disjoint triangles form a matching. Indeed, a vertex outside a triangle cannot be adjacent to two of its vertices: the edge joining them would have two common neighbours. Hence there are at most three edges between disjoint triangles.

Distinct triangles meet in at most one vertex. Consequently,

$$\sum_{m=0}^{3} N_m = \binom{231}{2} - 99\binom{7}{2} = 24486.$$

For an edge $uv$, each endpoint lies in six triangles other than the triangle containing $uv$. These two families give 36 disjoint pairs: an intersection would give another common neighbour of $u, v$. Counting edges between disjoint triangles yields

$$N_1 + 2N_2 + 3N_3 = 693 \cdot 36 = 24948. \tag{1}$$

Every pair of non-adjacent vertices has two common neighbours and therefore determines an induced four-cycle. A diagonal would have two common neighbours, contrary to $\lambda = 1$. Each four-cycle is counted by its two diagonals, so there are

$$\frac{1}{2} \cdot \frac{99 \cdot 84}{2} = 2079$$

such cycles. Complete a pair of opposite edges of a four-cycle to their triangles. Their third vertices are distinct, since otherwise a diagonal would have three common neighbours. The two triangles are disjoint. The two choices of opposite edges give precisely all pairs of edges between disjoint triangles. Thus

$$N_2 + 3N_3 = 2 \cdot 2079 = 4158, \qquad \delta = 3465 - 3N_3 \in 3\mathbb{Z}. \tag{2}$$

Combining the total pair count with (1) gives

$$N_0 = N_2 + 2N_3 - 462. \tag{3}$$

### 3.2 Two explicit tight frames

The common-neighbour counts give

$$A^2 = 12I - A + 2J, \qquad A\mathbf{1} = 14 \cdot \mathbf{1}.$$

On $\mathbf{1}^\perp$, the eigenvalues are $3, -4$. Their multiplicities $f, g$ satisfy $f + g = 98$ and $14 + 3f - 4g = 0$, so $f = 54$ and $g = 44$. Let $E$ be the orthogonal projection onto the $-4$ eigenspace. Then

$$G_0 := 63E = 27I - 9A + J, \qquad G_0^2 = 63G_0. \tag{4}$$

Choose vectors $w_u \in \mathcal{E} = \mathbb{R}^{44}$ with Gram matrix $G_0$ and put $y_u = w_u/3$. Let $W, Y$ be the matrices with these vectors as columns. Then

$$W^TW = G_0, \qquad WW^T = 63I, \qquad W\mathbf{1} = 0, \qquad WA = -4W, \tag{5}$$

$$Y^TY = 3I - A + J/9, \qquad YY^T = 7I, \qquad Y\mathbf{1} = 0, \qquad YA = -4Y. \tag{6}$$

These identities follow from the spectrum of $G_0$ and (4). In particular,

$$w_u^2 = 28, \qquad w_u \cdot w_v = \begin{cases} -8, & u \sim v, \\ 1, & u \not\sim v,\ u \neq v. \end{cases}$$

For a triangle $L = \{a, b, c\}$, define

$$x_L = y_a + y_b + y_c = (w_a + w_b + w_c)/3.$$

Its squared norm is $4$. Distinct intersecting triangles have inner product $0$: the Gram sum has four adjacency incidences and one diagonal term. Disjoint triangles with $m$ edges between them have inner product $1 - m$. Thus the Gram matrix $S = (x_L \cdot x_M)_{L,M}$ is integral with diagonal $4$.

Let $B$ be the vertex–triangle incidence matrix. Then $BB^T = 7I + A$, $X = WB/3$, and

$$XX^T = \frac{1}{9}W(7I + A)W^T = 21I, \qquad X\mathbf{1} = 0, \qquad \sum_{L \ni u} x_L = w_u. \tag{7}$$

The last identity follows from $(7w_u + \sum_{v\sim u} w_v)/3 = w_u$. Therefore

$$S^2 = 21S, \qquad S\mathbf{1} = 0, \qquad \Lambda := \sum_{L \in \mathcal{T}} \mathbb{Z}x_L$$

is an even integral lattice of rank $44$. To see discreteness, choose a real basis among the $x_L$. Its Gram matrix is invertible and integral, so the coordinates of all other generators have denominators bounded by its determinant. Evenness follows from the even diagonal of the Gram matrix.

### 3.3 The tensor and its trace identities

Set

$$T = \sum_{L \in \mathcal{T}} x_L^{\otimes 3}, \qquad U_v = \sum_L (v \cdot x_L) x_L x_L^T, \qquad Q(v) = \operatorname{tr}(U_v^2).$$

Tensor symmetry gives $U_v z = U_z v$, and the operators $U_v$ are self-adjoint. Write $Q(v) = v^T \mathcal{L} v$ for the resulting positive semidefinite operator $\mathcal{L}$. Contracting any two indices of $T$ gives $\sum_L x_L^2 x_L = 4\sum_L x_L = 0$, so $T$ is harmonic.

Its squared norm is

$$\begin{aligned} \|T\|^2 &= \sum_{L,M} (x_L \cdot x_M)^3 = 231 \cdot 64 + 2(N_0 - N_2 - 8N_3) \\ &= 13860 - 12N_3 = 4\delta. \end{aligned} \tag{8}$$

We have $\operatorname{tr}\mathcal{L} = \|T\|^2$. If

$$K_{LM} := \langle U_{x_L}, U_{x_M} \rangle_{\mathrm{HS}},$$

where the subscript denotes the Hilbert–Schmidt inner product, the tight frame (7) gives

$$K_{LL} = Q(x_L), \qquad \sum_L K_{LL} = 21 \operatorname{tr}\mathcal{L} = 84\delta. \tag{9}$$

**Lemma 3.1** (Diagonal congruence)**.** *For every triangle $L$, $K_{LL} \in 4\mathbb{Z}$.*

*Proof.* Index the triangles and write $s_{ij} = x_i \cdot x_j$. Expanding gives

$$K_{ii} = \sum_{j,k} s_{ij} s_{ik} s_{jk}^2.$$

Terms with $j = k$ are multiples of 16, and those with $j = i$ or $k = i$ are multiples of 4. Pairing the remaining terms gives

$$K_{ii} \equiv 2 \sum_{\substack{j<k \\ j,k \neq i}} s_{ij} s_{ik} s_{jk}^2 \pmod 4.$$

Modulo 2, replace $s_{jk}^2$ by $s_{jk}$. Now $(S^3)_{ii} = 21^2 s_{ii} = 1764$. The terms with a repeated index in this expression total

$$4^3 + 3 \cdot 4 \sum_{j \neq i} s_{ij}^2 = 64 + 12(84 - 16) = 880,$$

since $(S^2)_{ii} = 84$. Thus the sum over $j < k$, $j, k \neq i$, is $(1764 - 880)/2 = 442$, which is even. Hence $K_{ii} \equiv 0 \pmod 4$. □

# 4 A direct proof of $\delta \geq 165$

## 4.1 A congruence for the roots of $\Lambda$

**Lemma 4.1.** *If $r \in \Lambda$ and $r^2 = 2$, then $w_u \cdot r \in 3\mathbb{Z}$ for every $u$.*

*Proof.* For a vertex $u$ and a triangle $L$, the Gram matrix gives

$$w_u \cdot x_L = \begin{cases} 4, & u \in L, \\ -2, & u \notin L \text{ and } u \text{ has a neighbour in } L, \\ 1, & \text{otherwise.} \end{cases} \tag{10}$$

As already proved, an outside vertex has at most one neighbour in a triangle. All three values are congruent to 1 modulo 3. Since $r$ is an integral combination of the $x_L$, the integers $z_u = w_u \cdot r$ have a common residue modulo 3. Moreover, $\sum_u z_u = 0$ and $\sum_u z_u^2 = 63r^2 = 126$. If the residue is $\varepsilon \in \{1, -1\}$, every integer with that residue satisfies $z_u^2 + \varepsilon z_u \geq 2$. Summing would give $126 \geq 198$, a contradiction. The common residue is zero. □

## 4.2 Integral images and a norm gap

Fix $u$. The triangles containing $u$, those not containing $u$ but meeting its neighbourhood, and the remaining triangles number $7, 84, 140$, respectively. Write $s_0, s_1, s_2$ for their vector sums. By (7), $s_0 = w_u$. Summing triangles through neighbours of $u$ gives

$$-4w_u = \sum_{v \sim u} w_v = 2s_0 + s_1, \qquad s_0 + s_1 + s_2 = 0.$$

Hence $s_1 = -6w_u$ and $s_2 = 5w_u$. By (10),

$$U_{w_u} w_u = 16s_0 + 4s_1 + s_2 = -3w_u. \tag{11}$$

Put $M_u = -U_{w_u}/3$. This operator is self-adjoint and $M_u w_u = w_u$.

Write $r_u(L) = \mathbf{1}_{u \in L}$ and $n_u(L) = |N(u) \cap L|$. Then $w_u \cdot x_L = 1 + 9r_u(L) - 3n_u(L)$, and the tight frame gives

$$M_u = -7I - \sum_L (3r_u(L) - n_u(L)) x_L x_L^T. \tag{12}$$

By integrality of the Gram matrix, $M_u x_J \in \Lambda$ for all $u, J$. Self-adjointness and (11) imply

$$w_u \cdot M_u x_J = w_u \cdot x_J \in \{4, -2, 1\}.$$

The image is therefore nonzero, and Lemma 4.1 shows that it is not a root of $\Lambda$. Since the lattice is even and positive definite,

$$\|M_u x_J\|^2 \geq 4 \qquad \text{for every } u, J. \tag{13}$$

### 4.3 Rounding before summation

Tensor symmetry gives $M_u x_J = -U_{x_J} w_u/3$. For a fixed triangle $J$, the vertex frame implies

$$\sum_u \|M_u x_J\|^2 = \frac{1}{9}\,\mathrm{tr}\left(U_{x_J}^2 \sum_u w_u w_u^T\right) = 7K_{JJ}.$$

All 99 summands satisfy (13). Consequently,

$$7K_{JJ} \geq 396, \qquad K_{JJ} \geq 396/7 > 56.$$

Lemma 3.1 forces $K_{JJ} \geq 60$. Finally,

$$84\delta = \sum_J K_{JJ} \geq 231 \cdot 60 = 13860, \qquad \delta \geq 165.$$

Identity (2) then gives $N_3 \leq 1100$ and $N_2 \geq 858$. This argument does not use the lattice construction or any of the root bounds in the subsequent sections.

## 5 Construction of the auxiliary lattice

We now construct the lattice whose root rank will be bounded. This construction is not needed for the preceding bound on $\delta$.

### 5.1 A canonical even overlattice

Put $\Lambda_V = \sum_u \mathbb{Z} w_u$ and

$$N = \left\{ \frac{1}{3}\sum_u c_u w_u : c \in \mathbb{Z}^{99},\ \sum_u c_u \equiv 0 \pmod 3 \right\}. \tag{14}$$

Then $\Lambda_V \subseteq \Lambda \subseteq N \subseteq \frac{1}{3}\Lambda_V$. If $\sum c_u w_u = 0$, then $G_0 c = 0$ and $\sum c_u \equiv 0 \pmod 9$, because $G_0 = J + 9(3I - A)$. Thus the sum condition modulo 3 is well-defined on the represented vector, and

$$\left[\tfrac{1}{3}\Lambda_V : N\right] = 3.$$

For $s = \sum c_u$ and $s' = \sum c'_u$, both divisible by 3,

$$\left(\tfrac{1}{3}Wc\right)\cdot\left(\tfrac{1}{3}Wc'\right) = \frac{ss'}{9} + c^T(3I - A)c' \in \mathbb{Z}.$$

The squared norm has parity $(s/3)^2 + 3\sum c_u^2 - c^T A c \equiv s/3 + s \equiv 0 \pmod 2$. Hence $N$ is even and integral. Equivalently,

$$N = \Lambda_V + \sum_{u,v} \mathbb{Z}(y_u - y_v). \tag{15}$$

It follows that

$$f \in N^* \iff z = W^T f \in \mathbb{Z}^{99} \text{ and all } z_u \text{ are congruent modulo 3.} \tag{16}$$

For such $f$, we have $\sum z_u = 0$, $Az = -4z$, $z^2 = 63f^2$, and

$$21f = \tfrac{1}{3}Wz \in N.$$

Thus the discriminant group $N^*/N$ has exponent dividing 21.

### 5.2 The discriminant is elementary at seven

The common-residue condition in (16) eliminates the factor 3 directly. Write $z = r\mathbf{1} + 3a$, with $r \in \mathbb{Z}$ and $a \in \mathbb{Z}^{99}$. Since $\sum z_u = 0$, we have $\sum a_u = -33r$, divisible by 3. Also $Wz = WW^T f = 63f$ and $W\mathbf{1} = 0$, so

$$7f = \tfrac{1}{3}Wa \in N.$$

Thus $7N^* \subseteq N$. The discriminant group is a finite vector space over $\mathbb{F}_7$, and consequently

$$\det N = 7^e, \qquad N^*/N \simeq (\mathbb{Z}/7\mathbb{Z})^e. \tag{17}$$

### 5.3 Passing to determinant 49

For an even lattice $M$, its discriminant form is $q_M(v+M) = v^2 \bmod 2\mathbb{Z}$ on $M^*/M$. Milgram's formula [15, Appendix 4] states that

$$\frac{1}{\sqrt{|M^*/M|}} \sum_{d \in M^*/M} e^{\pi i q_M(d)} = e^{\pi i \operatorname{sign}(M)/4}. \tag{18}$$

A proof by Poisson summation in the positive-definite case is given in Appendix A. For $N$, whose rank is 44, the value is $-1$. Diagonalising its discriminant form over $\mathbb{F}_7$ expresses the normalised Gauss sum as a product of $e$ factors of the form $\pm i$. Indeed, the normalised quadratic Gauss sum over $\mathbb{F}_7$ is $i$, up to sign. Thus $e$ is even and nonzero.

A nondegenerate quadratic form over $\mathbb{F}_7$ in at least three variables has a nonzero isotropic vector: after diagonalisation, the sets $\{ax^2\}$ and $\{-c-by^2\}$ each have four elements in a set of seven, so they intersect. Complete such a vector to a hyperbolic plane and split it off orthogonally. Iterating leaves an anisotropic part of dimension at most two. Since $e$ is even, its dimension is zero or two. A hyperbolic plane has normalised Gauss sum 1, so the total value $-1$ excludes zero.

There is therefore a totally isotropic subspace $H \subset N^*/N$ of dimension $(e-2)/2$. Its preimage in $N^*$, denoted $\widehat{N}$, is an even integral lattice, and

$$N \subseteq \widehat{N} \subseteq N^*, \qquad [\widehat{N} : N] = 7^{(e-2)/2}, \qquad \det \widehat{N} = 49. \tag{19}$$

Evenness and integrality follow from $q_N|_H = 0$ and polarisation. The choice of $H$ need not be canonical. This is the classical isotropic-subgroup construction of overlattices [17].

*Remark* 5.1. The arguments of Sections 6–10 use only $N \subseteq M$ and the even integrality of $M$, not $\det M = 49$. For any such $M$, integrality implies $M \subseteq N^*$ and $M^* \subseteq N^*$. We may therefore write $\widehat{N} = M$ throughout those sections. In particular, their root-rank conclusion applies to every even integral overlattice of $N$.

## 6 The dual minimum and root geometry

### 6.1 The minimum of integral eigenvectors

**Lemma 6.1.** *If $0 \neq b \in \mathbb{Z}^{99}$ and $Ab = -4b$, then $b^2 \geq 14$. In the equality case, $b$ has seven coordinates equal to 1, seven equal to $-1$, and all remaining coordinates zero. Both supports are independent sets. The bipartite graph between them is 4-regular, and any two vertices on the same side have exactly two common neighbours on the other side.*

*Proof.* The coordinate sum is zero. Put $M = \max|b_u|$ and $t = \sum_{b_u>0} b_u$. Change the sign of $b$ so that its positive maximum is $M$. The corresponding row of $G_0 b = 63b$ gives

$$63M \leq 28M + (t-M) + 8t = 27M + 9t, \qquad \|b\|_1 = 2t \geq 8M.$$

If $M \geq 2$, then $b^2 \geq \|b\|_1 + M(M-1) \geq 18$. Thus, when $b^2 \leq 14$, every coordinate belongs to $\{0, 1, -1\}$. Let $m$ be the number of coordinates of each sign. The same inequality gives $m \geq 4$. Let $e_+, e_-$ be the numbers of edges within the two sign supports, $E = e_+ + e_-$, and $D$ the number of edges between them. Summing the eigenvector equation on each side gives $D = 4m + 2e_+ = 4m + 2e_-$. Hence $E$ is even and $D = 4m + E$.

Applying the positive semidefinite matrix $G_0$ to the indicator of the $2m$ supported vertices yields

$$0 \leq 4m^2 - 18m - 36E.$$

This is impossible for $m = 4$. For $m = 5, 6, 7$, the inequality and the evenness of $E$ force $E = 0$, so the bipartite graph is 4-regular. Counting common neighbours on the opposite side of positive pairs gives

$$m\binom{4}{2} \leq 2\binom{m}{2}, \qquad m \geq 7.$$

This excludes $b^2 < 14$. At $b^2 = 14$, we have $m = 7$ and equality in the last count. Therefore both allowed common neighbours of every same-side pair lie on the other side. □

For $0 \neq f \in N^*$, put $z = W^T f$. If its common residue modulo 3 is nonzero, the argument of Lemma 4.1 gives $z^2 \geq 198$ and $f^2 \geq 22/7$. If the residue is zero, write $z = 3b$; then $Ab = -4b$ and $f^2 = b^2/7 \geq 2$ by Lemma 6.1. Thus

$$\min N^* \geq 2, \qquad \min \widehat{N}^* \geq 2, \tag{20}$$

since $\widehat{N}^* \subseteq N^*$.

Every root $r \in \widehat{N}$ belongs to $\widehat{N}^* \subseteq N^*$, because $\widehat{N}$ is integral. Its code consequently satisfies

$$b = Y^T r \in \{0, 1, -1\}^{99}, \quad Ab = -4b, \quad b^2 = 14, \quad |P_r| = |N_r| = 7, \tag{21}$$

where $P_r, N_r$ are its positive and negative supports. Lemma 6.1 describes the biplane between these supports.

### 6.2 Root directions are orthogonal

Suppose roots $r, s$ have inner product $-1$. Then $r + s$ is also a root of $\widehat{N}$. Write their codes as $b, c, b + c$; each satisfies (21), and $b \cdot c = 7r \cdot s = -7$. A same-sign overlap would give a coordinate $\pm 2$ in $b + c$. Thus the supports overlap in seven vertices, with opposite signs, and their union has 21 vertices.

The six nonzero cells, labelled by $(b_u, c_u)$, are

$$X = (1, 0), \quad Y = (-1, 1), \quad Z = (0, -1), \quad -X, \ -Y, \ -Z.$$

The first three have a common size $m$, and their opposites have size $7 - m$. The sign supports of $b, c, b + c$ are independent. Consequently, the only allowed edges between these cells are those between $X, Y, Z$, those between their opposites, and those between the three pairs of opposite cells.

Two vertices in a cell $C$ have the same sign in two of the three codes. Both biplanes supply the two common neighbours allowed in the graph; the intersection of the relevant opposite sign supports is exactly $-C$. Thus all their common neighbours lie in $-C$. It follows that every allowed bipartite graph between non-opposite cells is a partial matching: a vertex adjacent to two vertices of one cell would supply a common neighbour outside its opposite cell.

For $u \in X$, let $x, y, d$ be its numbers of neighbours in $Y, Z, -X$. The two eigenvector equations give

$$-x - d = -4, \qquad x - y = 0.$$

Since $x \leq 1$, we have $d \geq 3$. The same argument applies to every cell. Choose a cell of size $L = \max(m, 7 - m) \geq 4$. If $L \geq 5$, its opposite has at most two vertices, a contradiction. If $L = 4$, all four vertices are adjacent to all three opposite vertices, giving three common neighbours to a non-adjacent pair. This is also impossible. Inner product $-1$ is excluded; replacing $s$ by $-s$ excludes inner product 1.

In an integral lattice, non-proportional roots have inner product $-1, 0$, or 1. Hence all root directions are orthogonal:

$$\Phi(\widehat{N}) = A_1^{\rho}. \tag{22}$$

### 6.3 The equitable partition of a root and active triangles

Fix a root $r$ and put $C = P_r \cup N_r$. A vertex outside $C$ cannot have two neighbours on the same side, since those two vertices already have both common neighbours on the opposite side. The equation $Ab = -4b$ forces it to have one neighbour on each side or none. Let $S$ be the vertices of the first kind and $I$ those of the second. Each of the 14 vertices in $C$ has ten outside neighbours, so $|S| = 70$ and $|I| = 15$. For the indicator vectors $c, s, i$,

$$Ac = 4c + 2s, \qquad As = 10c + 9s + 14i, \qquad Ai = 3s. \tag{23}$$

The second identity follows by applying $A^2 = 12I - A + 2J$ to $c$; the third follows from degree 14. Thus $I$ is independent and every vertex in $S$ has three neighbours in $I$.

The 28 edges between the two sign supports have 28 distinct third vertices, forming a set $B \subset S$. Put $F = S \setminus B$, of size 42. A triangle cannot contain two vertices of the same sign, so $a_L = r \cdot x_L \in \{0, 1, -1\}$. Call a triangle active when $a_L \neq 0$. Every active triangle contains one vertex of $C$ and two of $F$. Indeed, a vertex of $B$ already lies in the unique triangle on its edge to each of its two neighbours in $C$, whereas a vertex of $I$ has no neighbour in $C$. Each vertex of $C$ lies in three active triangles, and each vertex of $F$ in two. There are 21 active triangles of each sign, and

$$U_r r = 3y_C + 2y_F, \qquad y_E := \sum_{u \in E} y_u. \tag{24}$$

For any $r \in \widehat{N}$, the products $a_L$ are integral, and

$$U_r r - 21r = \sum_L (a_L^2 - a_L) x_L \in 2\Lambda.$$

Consequently,

$$t_r := \frac{U_r r - r}{2} \in \widehat{N}. \tag{25}$$

This is the only use of this tensor column in the root-rank bound.

# 7 Hoffman pairs and a first root-rank bound

## 7.1 Maximum independent sets

An independent set $H$ of size $h$ satisfies the Hoffman bound [13], already applied to this graph in [9, Theorem 4.3], $h \le 22$. Indeed, for $f = \mathbf{1}_H - (h/99)\mathbf{1}$, the least eigenvalue $-4$ gives

$$-14h^2/99 = f^T A f \ge -4(h - h^2/99), \qquad h \le 22.$$

At equality, $Af = -4f$, and

$$A\mathbf{1}_H = 4(\mathbf{1} - \mathbf{1}_H). \tag{26}$$

Every vertex outside $H$ therefore has four neighbours in $H$. We call such a set a Hoffman independent set. Its vector is

$$h_H = \tfrac{1}{7} y_H, \qquad h_H \cdot y_u = \mathbf{1}_H(u) - 2/9, \qquad h_H^2 = 22/9.$$

These formulas follow by applying the Gram matrix $Y^T Y$ to (26). For two such sets,

$$h_H \cdot h_{H'} = \frac{9|H \cap H'| - 44}{63}. \tag{27}$$

## 7.2 Two canonical Hoffman sets for each root

For a root $r$, partition (23) gives the independent sets of size 22

$$H_r^+ = P_r \cup I_r, \qquad H_r^- = N_r \cup I_r.$$

Applying $Y$ to (23) gives $y_{S_r} = -4y_{C_r}$ and $y_{I_r} = 3y_{C_r}$. The tight frame gives $y_{P_r} - y_{N_r} = 7r$. Therefore

$$h_r^{\pm} = \frac{y_{C_r} \pm r}{2}. \tag{28}$$

We need a precise integrality statement. Fix $\xi = y_0$. By (15), $y_u - \xi \in N \subseteq \widehat{N}$ and $3\xi = w_0 \in \widehat{N}$. Identities (24) and (25) give

$$h_r^- = t_r - y_{C_r} - y_{F_r}.$$

The sum $y_{C_r} + y_{F_r}$ has 56 terms and is congruent to $56\xi$ modulo $\widehat{N}$. Since $57\xi \in \widehat{N}$,

$$h_r^- \in \xi + \widehat{N}, \qquad h_r^+ = h_r^- + r \in \xi + \widehat{N}. \tag{29}$$

For every root $s$ of $\widehat{N}$, (21) gives $\xi \cdot s \in \mathbb{Z}$. Hence $h_r^{\pm} \cdot s$ is integral.

## 7.3 An integral Hoffman set cannot be active in two directions

Let $H$ be a Hoffman independent set such that every $h_H \cdot r_j$ is integral for the chosen roots. The tight frame gives

$$h_H \cdot r_j = \frac{|H \cap P_j| - |H \cap N_j|}{7} \in \{-1, 0, 1\}.$$

A nonzero product, after orienting the root, forces $P_j \subseteq H$ and $N_j \cap H = \varnothing$. Suppose that two directions $r_1, r_2$ are active and oriented this way. Opposite-sign intersections are empty, and orthogonality, $7r_1 \cdot r_2 = \sum_u b_1(u)b_2(u) = 0$, then forces $P_1 \cap P_2 = N_1 \cap N_2 = \varnothing$ as well.

Put $D = H \setminus (P_1 \cup P_2)$, of size 8, and $Z = V(G) \setminus (H \cup N_1 \cup N_2)$, of size 63. A vertex of $N_j$ already has four neighbours in $P_j \subset H$; by (26), it has no neighbour in $D$. All 14 neighbours of every vertex in $D$ therefore lie in $Z$. A vertex of $Z$ has at most one neighbour in each $P_j$, since two vertices of $P_j$ already have both common neighbours in $N_j$. The nexus 4 forces at least two neighbours in $D$. Counting edges between $D$ and $Z$ gives the impossible inequality

$$14 \cdot 8 \ge 2 \cdot 63.$$

Thus every $h_r^{\pm}$ is active only in its own root direction.

### 7.4 Orthogonal Hoffman sets and the Gram matrix of the centres

If $h_H \cdot r = 0$, then $|H \cap P_r| = |H \cap N_r| =: t$. We have $t \le 1$: two vertices of $P_r$ have six neighbours in total among the seven vertices of $N_r$, leaving at most one vertex of $N_r$ available for an independent set. Write $c = |H \cap C_r| \in \{0, 2\}$, $a = |H \cap S_r|$, and $d = |H \cap I_r|$. Counting edges between $C_r$ and $H$ using the nexus gives

$$4(14 - c) = 4c + 2a, \qquad a = 28 - 4c, \qquad d = 3c - 6.$$

Since $d \ge 0$, we must have $c = 2$. Therefore

$$|H \cap P_r| = |H \cap N_r| = 1, \qquad H \cap I_r = \varnothing. \tag{30}$$

For distinct directions $r_i, r_j$, all four intersections $H_i^{\varepsilon} \cap H_j^{\eta}$ consequently have size one. The sets $I_i$ are pairwise disjoint, avoid every $C_j$ for $j \ne i$, and

$$|P_i \cap P_j| = |P_i \cap N_j| = |N_i \cap P_j| = |N_i \cap N_j| = 1. \tag{31}$$

The centres $z_i = (h_i^+ + h_i^-)/2 = y_{C_i}/2$ have Gram matrix

$$(z_i \cdot z_j)_{i,j} = \frac{5}{2} I_\rho - \frac{5}{9} J_\rho.$$

Indeed, $y_{C_i}^2 = 3 \cdot 14 - 2 \cdot 28 + 14^2/9 = 70/9$, and products between distinct centres are $-5/9$ by (27). Positive semidefiniteness gives $5/2 - 5\rho/9 \ge 0$, whence

$$\rho \le 4. \tag{32}$$

## 8 Four directions leave only two cases

Suppose $\rho = 4$ and orient the roots $r_1, \dots, r_4$. The four sets $I_i$ occupy 60 vertices. Their complement $W$ has 39 vertices and contains all the supports $C_i$. Here and below $W$ denotes this vertex set rather than the frame matrix of Section 3. A vertex in $W$ has signature

$$b(v) = (r_1 \cdot y_v, \dots, r_4 \cdot y_v) \in \{0, 1, -1\}^4.$$

Let $n_a$ be the number of signatures of weight $a$, for $1 \le a \le 4$, and let $z$ count the zero signatures in $W$. Single incidences and the four cells for every pair of directions give

$$\sum_{a=1}^{4} a n_a = 56, \qquad \sum_{a=1}^{4} \binom{a}{2} n_a = 24, \qquad z = 7 - n_3 - 3n_4. \tag{33}$$

Figure 2 displays the partition and the local neighbour budget used next.

If $z > 0$, choose a zero-signature vertex $o$. It belongs to every $S_i$, so has three neighbours in each of the four disjoint sets $I_i$. These account for 12 neighbours. Its two remaining neighbours lie in $W$ and must carry the eight incidences required by one neighbour in each $P_i$ and each $N_i$. A vertex carries at most four such incidences. Both neighbours therefore have weight 4, giving $n_4 \ge 2$. The last identity of (33) now gives $z \le 1$. Thus

$$z \in \{0, 1\}. \tag{34}$$

When $z = 1$, the same identity forces $n_4 = 2$ and $n_3 = 0$. The other two moment identities then imply

$$z = 1 \implies (n_1, n_2, n_3, n_4) = (24, 12, 0, 2). \tag{35}$$

## 9 A weight obstruction to $z = 0$

Assume $z = 0$ and take $t \in I_k$. The root partition in direction $k$ gives no neighbours in $I_k$ or $C_k$. For each $j \ne k$, the disjointness properties proved in Section 7 give $t \in S_j$, so $t$ has three neighbours in $I_j$ and one in each of $P_j, N_j$. Consequently $t$ has nine neighbours in $\bigcup I_i$, and five in $W$. The total signature weight of those five neighbours is exactly six: there are two incidences in each of the other three directions and none in direction $k$.

Since $z = 0$, each vertex of $W$ has weight at least one. Five positive integer weights summing to six are at most two individually. Therefore no vertex of any $I_k$ is adjacent to a vertex of $W$ of weight at least three.

If a vertex $h$ has weight three, let $k$ be its missing direction. Then $h \in S_k$ and hence has three neighbours in $I_k$, a contradiction. Thus $n_3 = 0$. But (33) would now give $3n_4 = 7$, impossible for an integer $n_4$. We have proved

$$\rho = 4 \implies z \ne 0. \tag{36}$$

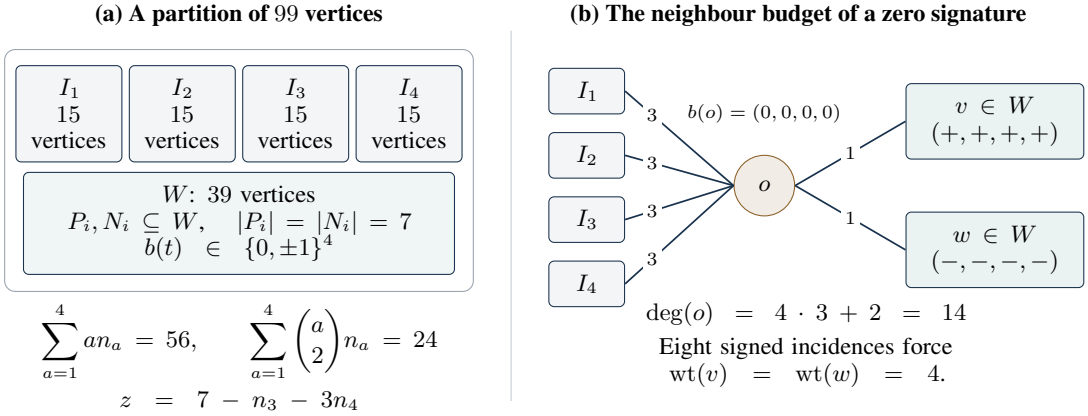



Figure 2: Conditional structure under $\rho = 4$, not a graph construction. In (a), $n_a$ counts weight-$a$ signatures in $W$, and $z$ counts zero signatures. In (b), a link labelled 3 records three neighbours in the indicated set $I_i$; only neighbours of $o$ are represented. The two remaining neighbours must together meet all eight signed supports $P_i, N_i$, forcing full, opposite signatures after orienting the roots.

## 10 Combinatorial exclusion of $z = 1$

This section does not use exhaustive search.

### 10.1 Vertices around the remaining point

Suppose $z = 1$. By (35), the remaining point $o$ has two neighbours $v, w$ of weight four and opposite signs in every direction. Orient the roots so that $v$ has all four positive signs and $w$ all four negative signs. A common neighbour of $v, w$ cannot belong to any $I_i$ or any $C_i$, since each sign support is independent. The only possibility is $o$, which is indeed adjacent to both. The parameters force $v, w$ to be adjacent, forming a triangle with $o$.

In each direction $v$ requires four neighbours of the opposite sign. The vertex $w$ accounts for four incidences in total. The other 12 neighbours of $v$, apart from $o, w$, must carry the remaining 12 incidences. Each belongs to at least one support, so all are singletons. The same holds for $w$. These two sets of 12 neighbours are disjoint and exhaust the 24 singleton vertices.

Let $V_i$ be the three neighbours of $v$ with signature $-i$, and $W_i$ the three neighbours of $w$ with signature $+i$. The same-sign cells for any pair of directions are occupied by $v, w$. The opposite-sign cells give the 12 vertices $R_{ij}$ with signature $+i, -j$, for $i \neq j$. Thus

$$P_i = \{v\} \cup W_i \cup \{R_{ij} : j \neq i\}, \qquad N_i = \{w\} \cup V_i \cup \{R_{ji} : j \neq i\}.$$

The neighbourhood of $v$, after removing the triangle $v, w, o$, forces a perfect matching within $V = \bigcup_i V_i$; the neighbourhood of $w$ forces one within $W = \bigcup_i W_i$. Each has six edges between distinct labels. The bipartite graph between $V$ and $W$ is also a perfect matching: a vertex of $V$ has exactly two common neighbours with $w$, namely $v$ and one vertex of $W$.

Each $R_{ij}$ has six neighbours in the sets $I_i$, none among $v, w, o$, two in $V$, and two in $W$, by $\mu = 2$ applied with $v, w$. It therefore has four neighbours among the other vertices $R$.

### 10.2 Six profiles saturate a twelve-edge budget

For each two-element set $D \subset \{1, 2, 3, 4\}$, put

$$s_D = y_v - \sum_{i \in D} r_i, \qquad s_D \cdot y_u = 1/9 + \ell_D(u).$$

The vertex Gram matrix and the signatures give

| $u$ | $\ell_D(u)$ |
|---|---|
| $v, w$ | $1$ |
| $o$ | $-1$ |
| $V_i$ | $-1 + \mathbf{1}_D(i)$ |
| $W_i$ | $-\mathbf{1}_D(i)$ |
| $R_{ij}$ | $-\mathbf{1}_D(i) + \mathbf{1}_D(j)$ |
| $\bigcup_i I_i$ | $0$ |

$$A\ell_D = -4\ell_D - 2 \cdot \mathbf{1}.$$

Each profile has six positive and seventeen negative coordinates. For a profile, let $P, B$ be these two supports, with $e_P, e_B$ their internal edge counts. The eigenvector equation yields

$$\deg_B(p) = \deg_P(p) + 6, \qquad \deg_P(b) = \deg_B(b) + 2, \qquad e_B = e_P + 1.$$

The number of common neighbours in $B$ of pairs in $P$ is at least

$$\sum_{b \in B} \binom{\deg_B(b) + 2}{2} \geq 17 + 4e_B = 21 + 4e_P.$$

Here $\binom{d+2}{2} \geq 1 + 2d$ for every nonnegative integer $d$. The entire graph permits only $30 - e_P$: each of the $\binom{6}{2}$ pairs has two common neighbours, except the $e_P$ edges, which have one. Thus $e_P \leq 1$. Since $vw$ is positive in all profiles, $e_P = 1$ and $e_B = 2$ in each of the six profiles.

Each edge of the internal matching on $V$ is negative in exactly one profile, and likewise for $W$. These 12 edges exhaust the budget $\sum_{|D|=2} e_{B_D} = 12$. No other edge can have both endpoints negative in any profile. The table therefore gives the restrictions

$$\begin{aligned} V_i\text{–}W_j &\implies i = j, \\ V_i\text{–}R_{jk} &\implies i = j, \\ W_i\text{–}R_{jk} &\implies i = k, \\ R_{ij}\text{–}R_{kl} &\implies j = k \text{ or } l = i. \end{aligned} \tag{37}$$

For example, if $i \neq j$, one can choose $D$ containing $j$ but not $i$, making both $V_i, W_j$ negative. For two arcs $R_{ij}, R_{kl}$, simultaneous negativity requires inclusion of their origins and exclusion of their destinations. This is impossible precisely when an origin is the other arc's destination.

### 10.3 The twelve-arc graph is forced

A vertex of $V_i$ has nine neighbours in the sets $I$, then $v$, its internal matching partner in $V$, its partner in $W$, and exactly two vertices $R_{ij}$. Each $R_{ij}$ has two neighbours among the three vertices of $V_i$. Thus the bipartite graph between $V_i$ and the three arcs originating at $i$ is the complement of a perfect matching. The same holds between $W_j$ and the arcs ending at $j$.

In the biplane of direction $i$, the vertex $R_{ij}$ has four neighbours in $N_i$: two in $V_i$, none at $w$, and hence two among the $R_{ki}$. Likewise it has two neighbours among the $R_{jk}$. The two possible sets overlap only at $R_{ji}$. Restriction (37) allows no other kind of $R$-neighbour, and its degree among $R$ is four. Thus $R_{ji}$ is not a neighbour, and all four other possibilities are neighbours:

$$N_R(R_{ij}) = \{R_{ki}, R_{jk} : k \notin \{i, j\}\}. \tag{38}$$

Two distinct arcs with the same origin have one common neighbour in $V_i$ and one in (38). They are non-adjacent and already have both common neighbours; no vertex in $I$ can be adjacent to both. The same holds for arcs with the same destination, using $W_j$. Two adjacent arcs $R_{ij}, R_{jk}$ with three distinct labels already have common neighbour $R_{ki}$, so no vertex in $I$ can be adjacent to both, by $\lambda = 1$.

### 10.4 Three common neighbours: the final contradiction

Let $t \in I_a \cap N(o)$. It has nine neighbours in the other sets $I$ and five outside their union. It is adjacent to neither $v$ nor $w$. The vertex $o$ is one of its two common neighbours with each of them, so it has exactly one neighbour in $V$ and one in $W$. Its two remaining neighbours outside the sets $I$, apart from $o$, are vertices $R$. Neither can involve label $a$, because $I_a$ avoids $C_a$. By the preceding paragraph, these two arcs have neither the same origin nor the same destination and are non-adjacent in (38). Among the six arcs on the other three labels, the only possible pairs are therefore $R_{ij}, R_{ji}$.

Fix an arc $R_{ij}$. It is non-adjacent to $o$. Their two common neighbours lie in $\bigcup_a (I_a \cap N(o))$, since $R_{ij}$ is adjacent to neither $v$ nor $w$. By the preceding paragraph, both are also adjacent to $R_{ji}$. These two distinct vertices consequently have three common neighbours, $o, R_{ij}, R_{ji}$, a contradiction. Hence

$$\rho = 4 \implies z \neq 1. \tag{39}$$

## 11 Completion of the proof

The counts in Section 3 give $\delta \in 3\mathbb{Z}$, and Section 4 gives $\delta \geq 165$. Section 5 constructs an even integral lattice $\widehat{N}$ of rank 44 and determinant 49 containing $\Lambda$. Section 6 proves that its dual minimum is at least 2 and that $\Phi(\widehat{N}) = A_1^\rho$. Section 7 gives $\rho \leq 4$. Under $\rho = 4$, Section 8 leaves only $z = 0$ or $z = 1$, excluded by Sections 9 and 10, respectively. Thus $\rho \leq 3$, completing the proof of Theorem 1.1.

All arguments used here are analytic, algebraic, or combinatorial; the proof requires no exhaustive enumeration. The two Lean formalizations prove the complete main theorem from the graph parameters alone. They sometimes use different, equivalent proofs; their scope and trust assumptions are specified in Appendix B.

## 12 Scope, limitations, and further questions

Theorem 1.1 imposes arithmetic and geometric restrictions on a hypothetical graph; it neither constructs a graph nor proves nonexistence. The numerical conclusions are

$$\delta \in 3\mathbb{Z}, \qquad 165 \leq \delta \leq 3465.$$

The upper bound here uses only $N_3 \geq 0$. We do not claim that every multiple of three in this interval survives other known obstructions, or that any such value is realizable.

The two main branches expose different obstacles. Strengthening the tensor estimate would require a further restriction on the local images or their simultaneous distribution; the trace identity alone does not grow more contradictory as $\delta$ increases. Root sparsity does not close that gap: $\rho = 0$ is allowed. Nor does the existence of an abstract lattice with the permitted determinant and minimum provide the 99 vertex vectors, 231 triangle vectors, and incidence relations required to recover a graph. A classification of such lattices without these distinguished configurations would not suffice.

These observations suggest precise questions rather than a claimed continuation of the proof. Can the local norm-gap images be coupled by a stronger integral moment identity? Can the distinguished vertex and triangle frames be excluded in a rootless determinant-49 extension? Can the coclique/profile budgets be formulated so as to control frame realizability even in the absence of roots? No answer to these questions is asserted here.

The present manuscript deliberately omits intermediate parameter exclusions and unsuccessful search branches that are not needed for Theorem 1.1. Its reproducibility claim concerns that theorem and the stated verification artifacts, not the entire research history.

## A Milgram's formula in the positive-definite case

For completeness we give the positive-definite specialization of the classical formula [15, Appendix 4]. See also Bailly–Cabral [16] for an elementary proof and discussion of its attribution. Let $M$ be an even positive-definite lattice of rank $n$, of covolume $v = \sqrt{\det M}$. Its discriminant group $D = M^*/M$ has $v^2$ elements. For $\operatorname{Im} z > 0$, put $\Theta_M(z) = \sum_{x \in M} e^{\pi i z x^2}$. Poisson summation for the Gaussian gives

$$\Theta_M(z) = v^{-1}(-iz)^{-n/2}\Theta_{M^*}(-1/z),$$

with the continuous branch taking positive real values on the positive imaginary axis. As $\varepsilon \downarrow 0$,

$$\varepsilon^{n/2}\Theta_M(i\varepsilon) \longrightarrow v^{-1}.$$

This is the Gaussian Riemann sum on $\sqrt{\varepsilon}M$; the same limit holds for every translate of $M$.

Since $M$ is even, $\Theta_M(1 + i\varepsilon) = \Theta_M(i\varepsilon)$. Moreover,

$$-\frac{1}{1+i\varepsilon} = -\frac{1}{1+\varepsilon^2} + i\frac{\varepsilon}{1+\varepsilon^2}.$$

On a coset $d + M$ of the dual, $x^2 \equiv q_M(d) \pmod{2\mathbb{Z}}$. The real correction to $-1$ is of order $\varepsilon^2$ and hence vanishes in the Riemann sum at scale $x = O(\varepsilon^{-1/2})$. More explicitly, after writing $t = \sqrt{\varepsilon}x$, the correction phase tends uniformly to 1 on compact sets, while for $0 < \varepsilon \le 1$ the modulus is bounded by $e^{-\pi\|t\|^2/2}$. Gaussian Riemann-sum tails are uniformly small: using a bounded fundamental cell, their sums are bounded by a constant times an integral of $e^{-c\|t\|^2}$ outside a ball of radius tending to infinity. This justifies the limit on every coset. Thus

$$\varepsilon^{n/2}\Theta_{M^*}\left(-\frac{1}{1+i\varepsilon}\right) \longrightarrow v^{-1}\sum_{d\in D} e^{-\pi i q_M(d)}.$$

Passing to the limit in Poisson summation, with $(-i)^{-n/2} = e^{\pi i n/4}$, gives

$$v^{-1} = v^{-2}e^{\pi i n/4}\sum_{d\in D} e^{-\pi i q_M(d)}.$$

Conjugating and multiplying by $v$ yields (18).

Here is also an explicit check of the Gauss-sum sign used over $\mathbb{F}_7$. Set $\zeta = e^{2\pi i/7}$ and $g = \sum_{t\in\mathbb{F}_7}\zeta^{t^2} = 1 + 2(\zeta + \zeta^2 + \zeta^4)$. The sum of all seven powers of $\zeta$ shows $\overline{g} = -g$. Also $|g|^2 = 7$: use the bijection $(x, y) \mapsto (x - y, x + y)$ and sum $\zeta^{(x-y)(x+y)}$ first in the second variable. Finally

$$\operatorname{Im} g = 2\big(\sin(2\pi/7) + \sin(4\pi/7) - \sin(\pi/7)\big) > 0.$$

The last inequality follows from $\sin(2\pi/7) > \sin(\pi/7) > 0$ and $\sin(4\pi/7) > 0$. Thus $g = i\sqrt{7}$. Multiplying the coefficient by a square leaves $g$ unchanged, and a nonsquare interchanges the nonzero squares and nonsquares, giving $-g$. For an order-7 discriminant class, $q$ takes values in $(2/7)\mathbb{Z}/2\mathbb{Z}$: $7q \in \mathbb{Z}$ by integrality and $49q \in 2\mathbb{Z}$ by evenness. Therefore each diagonal factor is exactly of this type.

## B Formal verification and reproducibility

### B.1 Statements and models

Two Lean 4 [20] developments using Mathlib [21] establish Theorem 1.1 from the strongly regular graph hypotheses alone. They were developed separately before the final cross-review, but share the informal mathematics. They are not independent discoveries of the theorem.

Both use vertex-indexed profiles in the $-4$ eigenspace, with pairing $z \cdot z'/63$. This realizes the Euclidean representation of Section 3. The formal conclusions construct an actual integral basis of size 44 and prove that its Gram determinant is 49; the rank assertion is not a statement quantified over possibly nonexistent bases. The root bound is expressed either as a bound on the span or as the impossibility of four roots distinct modulo sign; the proved orthogonality makes these equivalent.

The final entry points are

| Development | Lean | Final theorem |
|---|---|---|
| Codex | `v4.24.0` | `Conway99.graph_minimal_article_theorem` |
| Opus | `v4.34.1` | `Conway99.theorem_1_1` |

The pinned Mathlib revisions are:

| | |
|---|---|
| Codex | `f897ebcf72cd16f89ab4577d0c826cd14afaafc7` |
| Opus | `d13f23b723b8a846827a245b89c10fc7d3f11612` |

The projects must be built separately rather than merged or upgraded.

The developments need not follow every displayed proof line. In particular, Codex retains a kernel-checked finite certificate for the zero-signature case, whereas Opus proves the short weight argument of Section 9. Both close the Milgram step: no unproved Poisson or Gauss-sum interface remains as a hypothesis of either final theorem. The hexagon remark, editorial novelty comparisons, and research history are not formalized.

### B.2 Checks and trust boundary

The completed builds and axiom inspections are recorded with the sources. The final theorems report only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no admitted proofs or additional axioms in their dependencies. The verification policy rejects `sorry`, `admit`, native proof computation, and kernel-bypass mechanisms.

After completion, each development received a targeted cross-review by the other AI system. The reviews checked final statements, mathematical modelling, proof-source integrity, builds, and critical interfaces, including the dual, exact Gram pairing, non-vacuous rank, determinant descent, and the Gauss-sum sign. They are AI-generated technical assessments, not human peer review or an absolute correctness guarantee. Recorded builds reused dependency and import caches; they must not be represented as fresh cold rebuilds. Source hashes establish identity, not mathematical correctness. The trust boundary includes the Lean kernel and the correspondence between the formal statements and their mathematical interpretation.

### B.3 Companion repository

The formal sources and verification materials are available at:

`https://github.com/Math680/conway99-minimal-lean`

The repository contains separate `codex/` and `opus/` Lake projects, toolchain pins and dependency locks, a mathematical proof map, source hashes, build instructions, and both cross-review statements. After installing the selected toolchains and obtaining their Mathlib caches, the reproduction commands from its root are

```
python3 scripts/verify.py --project codex
python3 scripts/verify.py --project opus
```

On Windows, `python` may replace `python3`. The source-only option checks integrity and policy, not proofs. The Codex packing-certificate branch can require substantial memory (roughly 20 GB in an earlier local run). An out-of-memory failure is neither a proof check nor a mathematical contradiction. The short combinatorial route is contained in the Opus project.

## C AI research protocol and human contribution

### C.1 Scope of autonomy

The research was not limited to AI copy-editing or isolated suggestions. Agents proposed mathematical strategies, derived and revised lemmas, searched the literature, wrote exploratory programs and exact certificates, and developed the two Lean formalizations. The human author, Mathieu Rannou, initiated the project, designed and steered its use of AI systems, selected priorities, allocated resources, and relayed results between systems. He did not personally derive every proof and does not claim comprehensive personal mastery of all technical arguments. Attribution of authorship must not be read as a claim of independent human re-derivation of the mathematics. The author is responsible for the decision to disseminate the work and for representing this provenance transparently.

### C.2 Systems and chronology

The following operational account uses the model labels and chronology reported by the project organizer. It describes roles in this project, not externally benchmarked capabilities or an independently audited inventory of provider versions. Research began in July 2026 with GPT-5.6 Sol and its subagents. GPT-6 Sol was also used briefly before the release of GPT-6.1 Sol. It resumed in early October 2026 with GPT-6.1 Sol as the principal Codex orchestrator, using its own subagents and, for selected tasks, Gemini 3.8 Flash agents through Antigravity. Claude Opus 5.5 carried out substantial complementary investigations. Codex designed and issued the delegated research tasks: it invoked Antigravity directly through the CLI and prepared written prompts for Opus, which the human organizer relayed along with the resulting handoffs. The human also adjusted priorities and initiated selected complementary investigations.

Human–agent exchanges were conducted primarily in French, while mathematical notes and inter-agent handoffs used both French and English. The manuscript was prepared in English.

Tasks included both targeted proof questions and broader attempts at a global obstruction. Exploratory outputs were not automatically treated as theorems: arithmetic compatibility, floating-point evidence, conditional lemmas, exact finite certificates, and completed proofs were distinguished. Only the dependency chain supporting the main theorem is presented in this paper.

Once the informal proof chain retained for the article was established, Codex and Opus undertook separate Lean formalizations of those results. Only after both developments were completed did they cross-review each other's formal sources and theorem interfaces. Figure 3 distinguishes these stages from the preceding research orchestration.

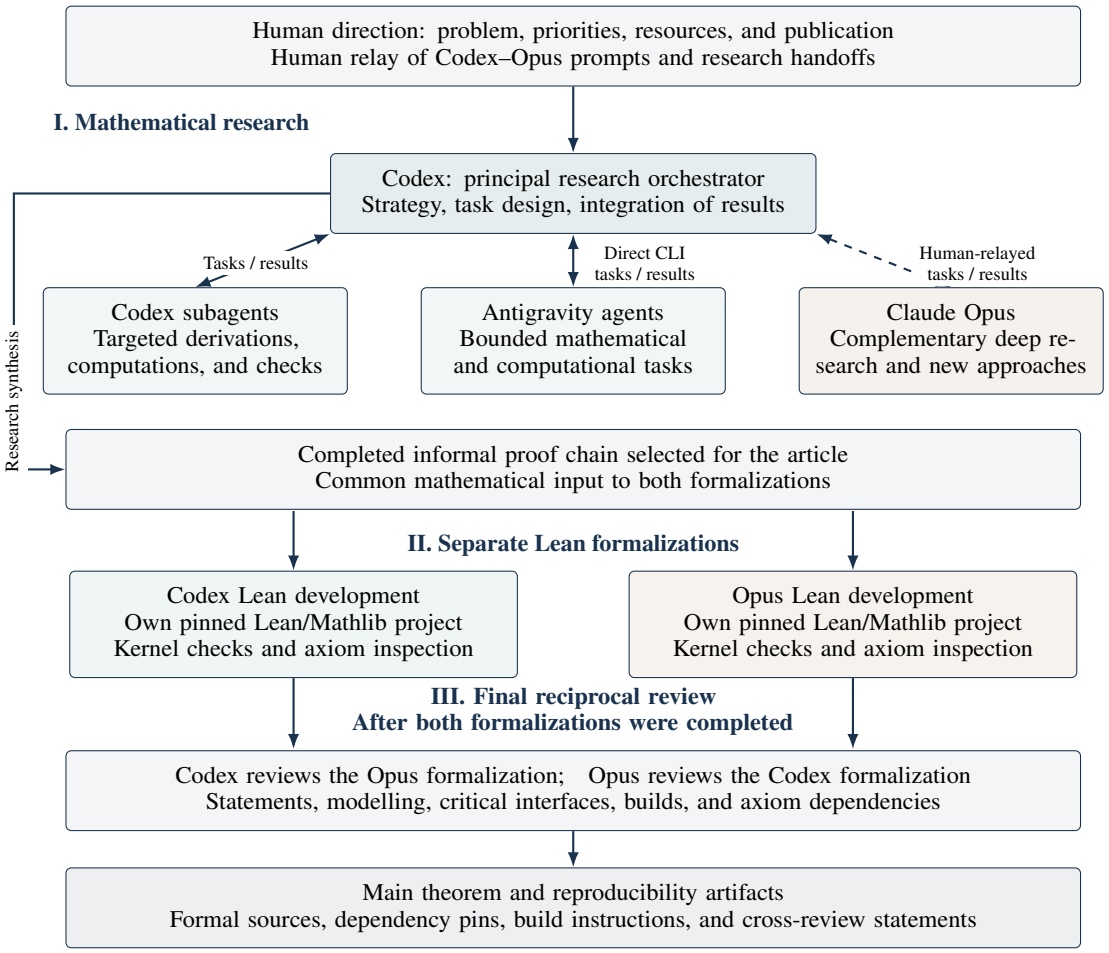



Figure 3: Research orchestration and subsequent verification workflow. Codex delegated tasks directly to its subagents and Antigravity, and prepared tasks for Opus that were relayed by the human organizer. Research handoffs returned to Codex for integration. The resulting informal proof chain was then formalized separately by Codex and Opus, before the final reciprocal review. This is an operational chronology, not a claim of statistical independence or human peer-review certification.

### C.3 Recorded token usage

Table 1 summarizes the local usage records inspected on 16 October 2026. Counts include repeated context processing and cache reads; they do not count only newly written text. Codex totals exclude duplicated events and inherited subagent counter baselines. Antigravity calls and Opus response blocks were deduplicated by request identifiers. The figures are a fixed, rounded snapshot, not live counters.

The Codex record starts in July 2026; the Antigravity and Opus records cover their October work, with Opus beginning on 5 October. The Antigravity total excludes the human-initiated report conversation and a separately identified, apparently manual audit of $\delta = 33$. The Opus total excludes a small auxiliary Haiku usage record. Reasoning tokens are already included in output totals. The systems use provider-specific token accounting, so this sum is a bookkeeping summary, not a common measure of computation, financial cost, or mathematical contribution. It is not a billing or quota audit.

| System | Included scope | Tokens ($10^6$) |
|---|---|---|
| Codex | Main conversation and 64 subagent sessions | 1,235.1 |
| Antigravity | 21 delegated research conversations | 42.3 |
| Claude Opus | Research, Lean formalization, and cross-review | 674.8 |
| Total | | 1,952.2 |

Table 1: Recorded project token usage, including cache reads and reasoning output. The sum is approximately 1.95 billion tokens.

### C.4 Validation and limitations

Validation combined explicit mathematical derivations, exact arithmetic checks of finite statements, targeted review of delicate steps, and kernel checking of the final Lean developments. Formalization also simplified the informal presentation, notably the common-residue argument for $7N^* \subseteq N$ and the weight obstruction for $z = 0$. The protocol did not require every exploratory calculation to be reproduced independently. It did require closing the main theorem without remaining auxiliary hypotheses and documenting the formal trust boundary.

AI systems can share training data, mathematical conventions, and failure modes. Two agreeing agents therefore do not constitute two independent human referees. Likewise, compilation does not verify priority, exposition, or the semantic interpretation of a badly chosen statement. Those limitations motivate the explicit statement reviews, references, and scope disclosures in this manuscript. No claim of complete experimental reproducibility of every historical conversation is made; the reproducible objects are the mathematical proofs, pinned formal sources, and recorded verification procedures.

## Acknowledgements

The author acknowledges the prior work cited throughout the paper and the developers and contributors of Lean and Mathlib. The AI systems and their substantive research roles are disclosed in Appendix C. Their technical cross-reviews should not be construed as endorsements by their providers.

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